$2^n > n^2 + n$

Use mathematical induction to show that whenever n is an integer greater than 4.

Soln: Let $p(n): 2^n > n^2 + n$

$p(5): 2^5 > 5^2 + 5$

$32 > 30 \Rightarrow$ True.

$\therefore p(n)$ is true for $n = 5$.

Assume that $p(n)$ is true for $n = m$.

$$2^m > m^2 + m \quad \text{——— (i)}$$

Now, we have to show that, $p(n)$ is true for $n = m+1$.

we have to show that $2^{m+1} > (m+1)^2 + (m+1)$

——— (ii)

multiply both sides of (i) by 2

$2^m \cdot 2 > 2m^2 + 2m$

$2^{m+1} > 2m^2 + 2m$

$2^{m+1} \geq 2m^2 + m + m$

$2^{m+1} > 2m^2 + m + 1$

$2^{m+1} \geq m^2 + m^2 + m + 1$

$2^{m+1} > m^2 + 2m + 1 + m + 1$

$2^{m+1} \geq (m+1)^2 + (m+1)$.

$\therefore 2^{m+1} > (m+1)^2 + (m+1)$

$\therefore$ $p(n)$ is true for $\forall n > 4$. (proved).

Use mathematical!' induction to show that ~~$1 \cdot 20 + 2 \cdot 21 + 3 \cdot 22 + \cdots + n \cdot 2n \cdot 1 = (n-1) \cdot 2n$ $+1$~~ whenever n is a positive integer.

$$1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \cdots + n \cdot 2^{n-1} = (n-1) \cdot 2^n + 1 .$$

Soln:

$$p(n): 1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \cdots + n \cdot 2^{n-1} = (n-1) \cdot 2^n + 1$$

$p(1)$: $1 \cdot 2^0 = (1-1)\, 2^1 + 1$

$\Rightarrow 1 = 1$

$\therefore p(1)$ is true.

$p(2)$:

$1 \cdot 2^{1-1} + 2 \cdot 2^{2-1} = (2-1) \cdot 2^2 + 1$

Assume that, $p(n)$ is true for $n = m$

$$1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \cdots + m \cdot 2^{m-1} = (m-1) \cdot 2^m + 1 \quad \text{— (i)}$$

Now we have to show that $p(n)$ is true for $n = m+1$

$$1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \cdots + m \cdot 2^{m-1} + (m+1) \cdot 2^{(m+1)-1} = (m+1-1) \cdot 2^{m+1} + 1$$

$$= m \cdot 2^{m+1} + 1$$

Add $(m+1) \cdot 2^{m+1-1}$ on both sides of (i). — (ii)

$$1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \cdots + m \cdot 2^{m-1} + (m+1) \cdot 2^{(m+1)-1} = (m-1) \cdot 2^m + 1 + (m+1) \cdot 2^{m+1-1}$$

$$= (m-1)\, 2^m + 1 + (m+1) \cdot 2^m$$

$$= 2^m (m-1+m+1) + 1$$

$$= m \cdot 2 \cdot 2^m + 1 \quad = m \cdot 2^{m+1} + 1$$

$\therefore$ $p(n)$ is true for $\forall\, n$. (proved).

3. Use mathematical induction to show that

$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6} n(n+1)(2n+1)$ when n is a positive integer.

Soln: Let $p(n)$: $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6} n(n+1)(2n+1)$

$p(1)$: $1^2 = \frac{1}{6} 1 \cdot (2) \cdot (3)$

$1 = 1$

$\therefore$ $p(1)$ is true.

$p(2)$:

L.H.S $= 1^2 + 2^2 = 5$

R.H.S $= \frac{1}{6} \cdot 2 \cdot 3 \cdot 5 = 5$

$\therefore$ L.H.S = R.H.S

$\therefore$ $p(2)$ is true

Assume that, $p(n)$ is true for $n = m$.

$1^2 + 2^2 + 3^2 + \cdots + m^2 = \frac{1}{6} m(m+1)(2m+1)$ ——— (i)

Now we have to show that, $p(n)$ is true for $n = m+1$.

that is. $1^2 + 2^2 + 3^2 + \cdots + m^2 + (m+1)^2 = \frac{1}{6}(m+1)(m+1+1)(2(m+1)+1)$

$= \frac{1}{6}(m+1)(m+2)(2m+3)$ ——— (ii)

Now add $(m+1)^2$ on both sides of (i)

$1^2 + 2^2 + 3^2 + \cdots + m^2 + (m+1)^2 = \frac{1}{6} m(m+1)(2m+1) + (m+1)^2$

$= (m+1)\left(\frac{1}{6} m(2m+1) + (m+1)\right)$

$= (m+1)\ \frac{m(2m+1) + 6(m+1)}{6}$

$= (m+1)\ \frac{2m^2 + m + 6m + 6}{6}$

$= \frac{1}{6}(m+1)\ (2m^2 + 7m + 6)$

$= \frac{1}{6}(m+1)\ (2m^2 + 4m + 3m + 6)$

$$= \frac{1}{6}(m+1)\left(2m(m+2)+3(m+2)\right)$$

$$= \frac{1}{6}(m+1)(m+2)(2m+3)$$

$\therefore P(n)$ is true for $\forall n \geq 1$.

Graph theory

Relations:

let A be a non-empty set.

Then Relation. $R \subseteq A \times A$

if A & B are two non-empty sets

then relation R from A to B is denoted by $\boxed{R \subseteq A \times B}$

Example: If $A = \{1, 2, 3, 4\}$

$B = \{3, 5, 8, 6\}$

(i) Find a Relation $R = \{(a,b) : a \in A, b \in B \text{ and } a+b \geq 10\}$

Soln: $A \times B = \{(1,3), (1,5), (1,8), (1,6), (2,3), (2,5), (2,8), (2,6), (3,3), (3,5), (3,6), (3,8), (4,3), (4,5), (4,6), (4,8)\}$

$R = \{(2,8), (3,8), (4,6), (4,8)\}$

Ans

(ii) $R = \{(a,b) : a \in A, b \in B \text{ and } a > b+1\}$

$R = \{\}$ or $R = \phi$.

Ans

(iii) $R = \{(a,b) : a \in A, b \in B \text{ and } b - a \geq 2\}$

$= \{(1,3), (1,5), (1,8), (1,6), (2,5), (2,8), (2,6), (3,5)$

$(3,6), (3,8), (4,6), (4,8)\}$

Ans

4 - types of Relation:

(i) Reflexive . $R = \{(a,b) : a \in A, b \in B \text{ and } a = b\}$

(ii) Symmetric . $R = \{(a,b) : a \in A, b \in B, \text{ if } (a,b) \text{ then } (b,a)$ must be present$\}$

(iii) Anti symmetric:

$R = \{(a,b) \ldots\}$

if $(a,b) \in R$

$(b,a) \notin R$

(iv) Transitive Relation:



(a,b) & (b,c)

(a,c)

For each of these relations on the set {1,2,3,4}, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

a) {(2,2),(2,3),(2,4),(3,2),(3,3),(3,4)}
b) {(1,1),(1,2),(2,1),(2,2),(3,3),(4,4)}
c) {(2,4),(4,2)}
d) {(1,2),(2,3),(3,4)}
e) {(1,1),(2,2),(3,3),(4,4)}
f) {(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)}

Soln: (a) $R = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

$1R_1$ or $(1,1) \notin R$

$4R_4$ or $(4,4) \notin R$

$\therefore$ R is not reflexive.

Symmetric:

$2R_3 \rightarrow 3R_2$

$2R_4 \rightarrow 4R_2$

not symmetric.

Anti-Symmetric:

$2R3$ $3R2$

$\therefore$ not Anti-symmetric

Transitive

$(2,2) \rightarrow (2,3) \rightarrow (2,3)$ ✓

$(2,4) \rightarrow (2,4)$ ✓

$(2,3) \rightarrow (3,2) \rightarrow (2,2)$ ✓

$(3,3) \rightarrow (2,3)$ ✓

$(3,4) \rightarrow (2,4)$ ✓

$(2,4) \rightarrow$ ✓

$(3,2) \rightarrow (2,2) \rightarrow (3,2)$ ✓

$(2,3) \rightarrow (3,3)$ ✓

$(2,4) \rightarrow (3,4)$ ✓

$(3,3) \rightarrow (3,2) \rightarrow (3,2)$ ✓

$\rightarrow (3,4) \rightarrow (3,4)$ ✓

$(3,4) \rightarrow$

$\therefore$ R is a transitive.

(ii) $R = \{ (1,1), (1,2), (2,1), (2,2), (3,3), (4,4) \}$

Reflexive: R is a Reflexive relation : (1,1)
(2,2)
(3,3)
(4,4)

Symmetric: R is symmetric

Anti-symmetric: 1 R 2 but 2 R 1

∴ not anti symmetric.

Transitive:

(a,b)	(b,c)	(a,c)
→ (1,1)	(1,2)	(1,2) ✓
→ (1,2)	(2,1)	(1,1) ✓
	(2,2)	(1,2) ✓

(2,1)	(1,1)	→	(2,1) ✓
	(1,2)	→	(2,2) ✓
(2,2)	(2,1)	→	(2,1) ✓
	(2,2)	→	(2,2) ✓
(3,3)	→		✓
(4,4)	→		✓

∴ Transitive.

© $R = \{(2,4), (4,2)\}$

$(2,4) \quad (4,2) \longrightarrow (2,2)$ ✗

R is not reflexive, Anti-symmetric, transitve

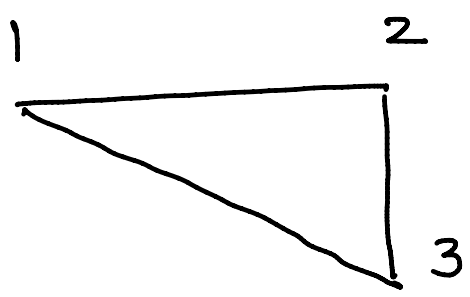
It's only symmetric.

(d) $R = \{(1,2), (2,3), (3,4)\}$

R is anti-symmetric

$(1,2) \longrightarrow (2,3) \longrightarrow (1,3)$ ✗

∴ R is not transitve, symmetric & Reflexive.



(f) $R = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$

not Reflexive

not Symmetric.

IR4 ← not Symmetric.

4R1

not Anti-symmetric

(1,3) (3,1)

(1,3) (3,1)

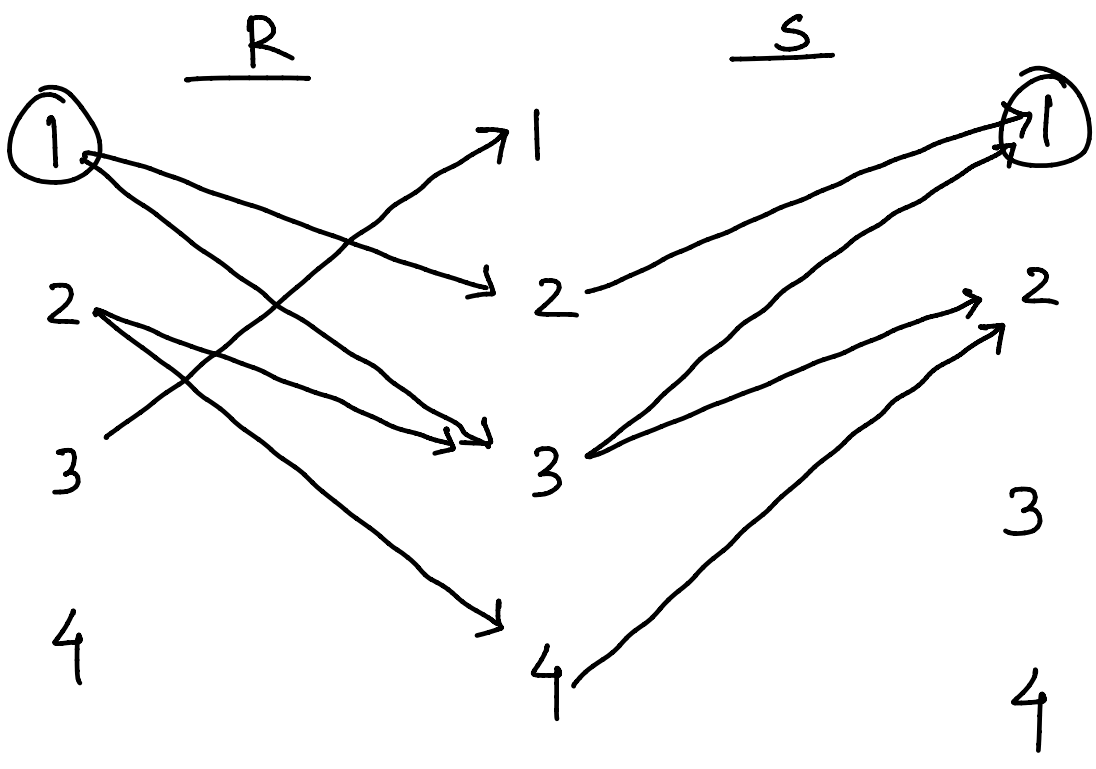
(3,4)

(1,1) ✗

(1,4) ✓

∴ not transitive.

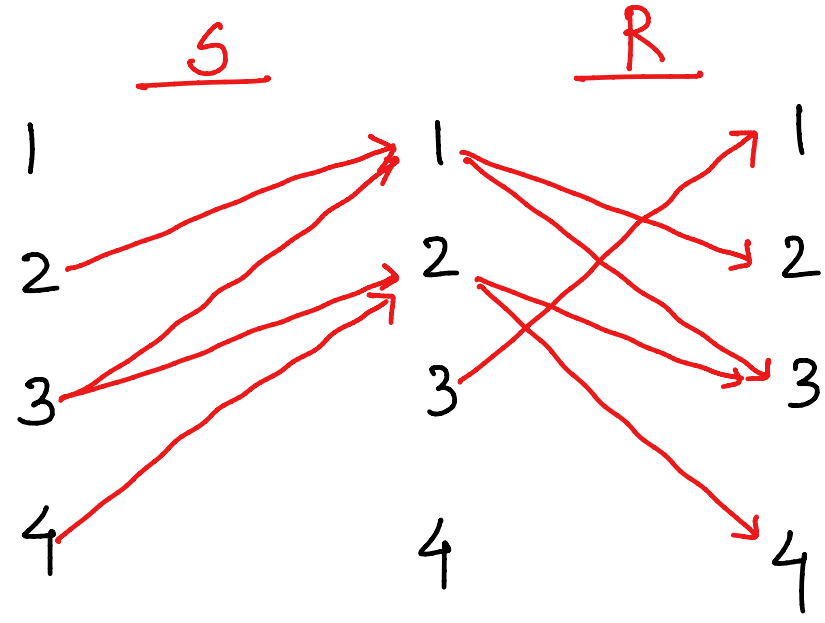
Let R be the relation {(1,2),(1,3),(2,3),(2,4),(3,1)}, and let S be the relation {(2,1),(3,1),(3,2), (4,2)}.Find S ◦R.



$SoR = \{(1,1), (1,2), (2,1), (2,2)\}$

Ans

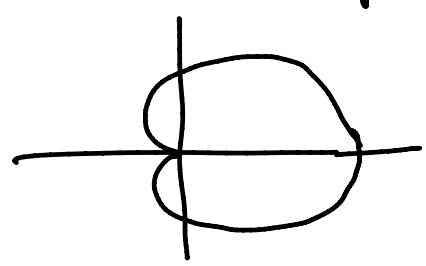
Find ROS



$ROS = \{(2,2), (2,3), (3,2), (3,3), (3,4), (4,3), (4,4)\}$

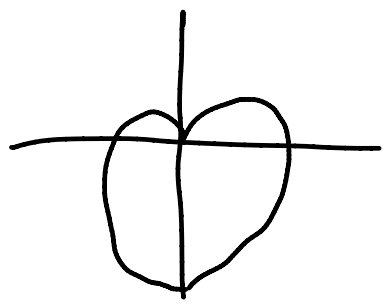
Ans

$r = a(1+\cos\theta)$

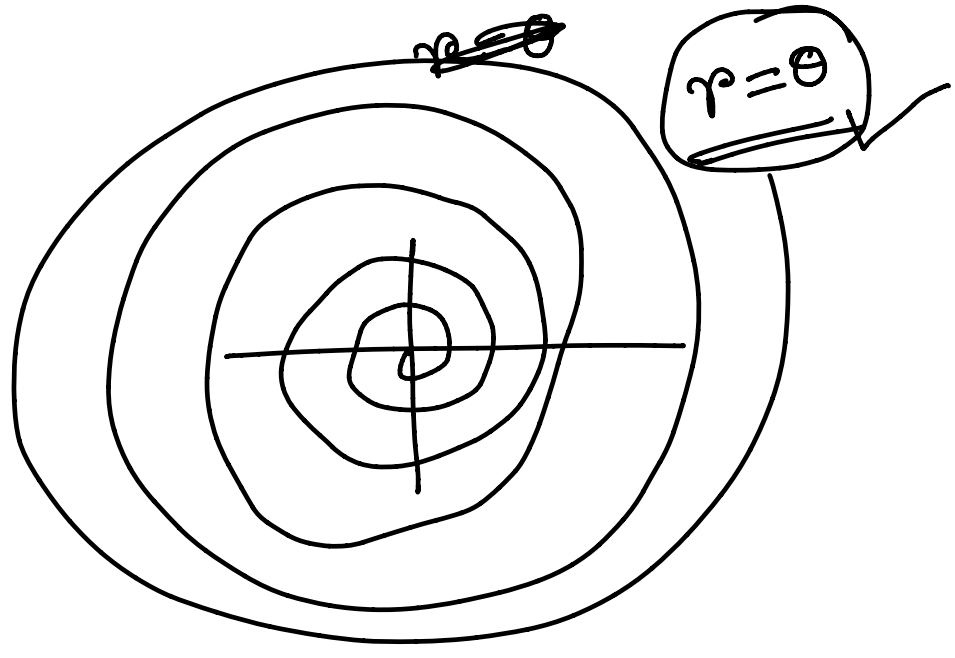


$r = 1$

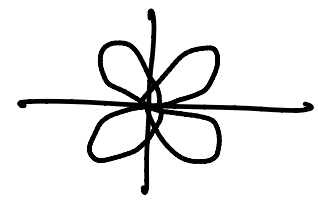
$\theta = 1$



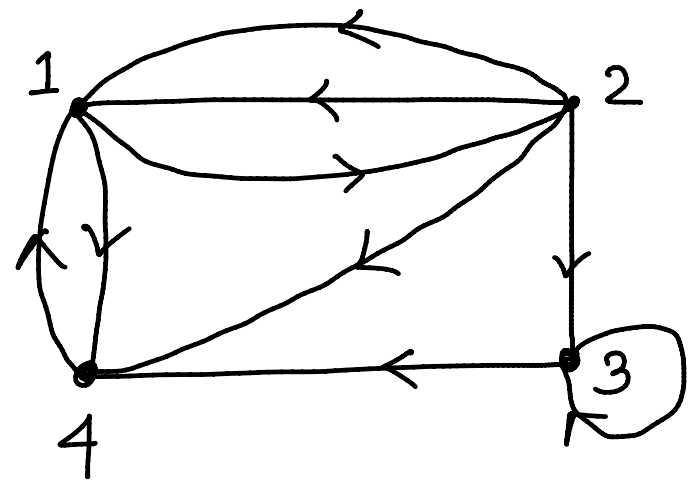
$r = a$



$r = \sin 2\theta$



Relations: (R)



$$R = \{(1,2), (1,4), (2,1), (2,3), (2,4), (3,3), (3,4), (4,1)\}$$

→ If A is a non-empty set. then $R \subseteq A \times A$

$$R = \{(a,b) : a \in A,\ b \in A \text{ and some conditions}\}$$

→ If A & B two non-empty set. then $R \subseteq A \times B$

$$R = \{(a,b) : a \in A,\ b \in B \text{ and some conditions}\}$$

Example: Find R on $A = \{1,2,4,6\}$. $B = \{4,5,6\}$

(i) with $R = \{(a,b) : a \in A,\ b \in B \text{ and } a+b \geq 8\}$

Soln:

$$A \times B = \{(1,4), (1,5), (1,6), (2,4), (2,5), (2,6), (4,4), (4,5), (4,6), (6,4), (6,5), (6,6)\}$$

$$R = \{(2,6), (4,4), (4,5), (4,6), (6,4), (6,5), (6,6)\}$$

Ans

(ii) $R = \{(a,b) : a \in A,\ b \in B \text{ and } a+2 \geq b\}$

$$= \{(2,4), (4,4), (4,5), (4,6), (6,4), (6,5), (6,6)\}$$

4-types of Relation:

i. Reflexive : (1,1), (2,2), (3,3). (a,a)

$R = \{(a,b) : a \in A, b \in B \text{ and } a = b\}$

(ii) Symmetric : if (a,b) is present then (b,a) must be present.

(iii) Antisymmetric : if (a,b) is present then (b,a) not.

(iv) Transitive : If (a,b) & (b,c) is present then (a,c) must be present.

For each of these relations on the set {1,2,3,4}, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

a) {(2,2),(2,3),(2,4),(3,2),(3,3),(3,4)}
b) {(1,1),(1,2),(2,1),(2,2),(3,3),(4,4)}
c) {(2,4),(4,2)}
d) {(1,2),(2,3),(3,4)}
e) {(1,1),(2,2),(3,3),(4,4)}
f) {(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)}

$R \subseteq A \times A$

$A = \{1,2,3,4\}$

(a) $R = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

(i) $(1,1) \notin R$ ∴ R is not reflexive.

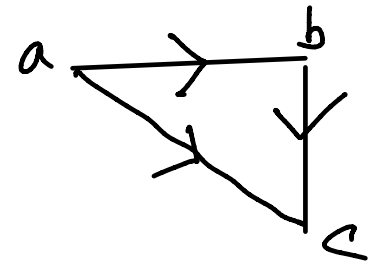
(ii) $(2,4) \in R$ but $(4,2) \notin R$ ∴ R is not symmetric.

(iii) $(2,3) \in R$, $(3,2) \in R$ } R is not Anti-symmetric.

(iv)

(2,2)	(2,2)	(2,2)
	(2,3)	(2,3)
	(2,4)	(2,4)
(2,3)	(3,2)	(2,2)
	(3,3)	(3,3)
	(3,4)	(2,4)
(2,4)		
(3,2)	(2,2)	(3,2)
	(2,3)	(3,3)
	(2,4)	(3,4)
(3,3)	(3,2)	(3,2)
	(3,3)	(3,3) ✓
	(3,4)	(3,4)
(3,4)		

$\therefore$ Transitive.



$a \to b$

(b) Reflexive ✓

Symmetric ✓

$(1,2) \in R$

$(2,1) \in R$

$\therefore$ R is not Anti-symmetric.

$R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$

(1,1) — (1,1) → (1,1) ✓

(1,2) → (1,2)

(1,2) (2,1) — (1,1) ✓

(2,2) (1,2)

(2,1) (1,1) → (2,1) ✓

(1,2) (2,2)

(2,2) (2,1) → (2,1) ✓

(2,2) (2,2)

(3,3)

(4,4)

$\therefore$ R is transitive.

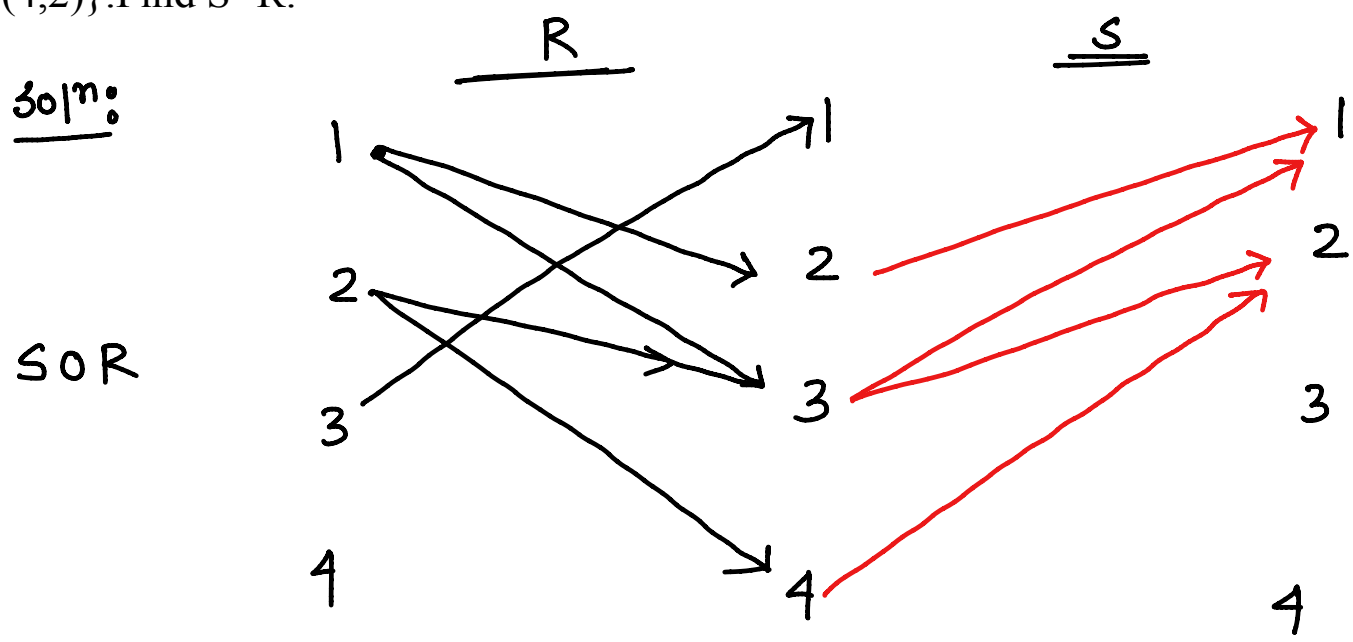
(c) $R = \{(2,4), (4,2)\}$

(2,4) (4,2) (2,2) ✗

not transitive.

Let R be the relation {(1,2),(1,3),(2,3),(2,4),(3,1)}, and let S be the relation {(2,1),(3,1),(3,2),(4,2)}.Find S ◦R.

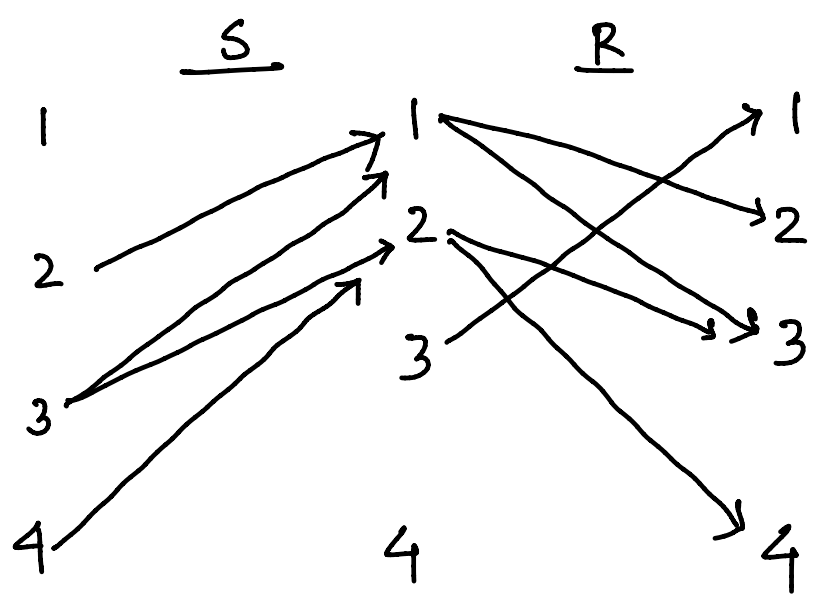
Soln:



$$SOR = \{(1,1), (1,2), (2,1), (2,2)\}$$

Ans

ROS:



$$ROS = \{(2,2), (2,3), (3,2), (3,3), (3,4), (4,3), (4,4)\}$$

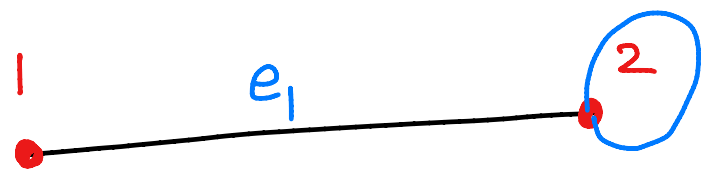
Ans

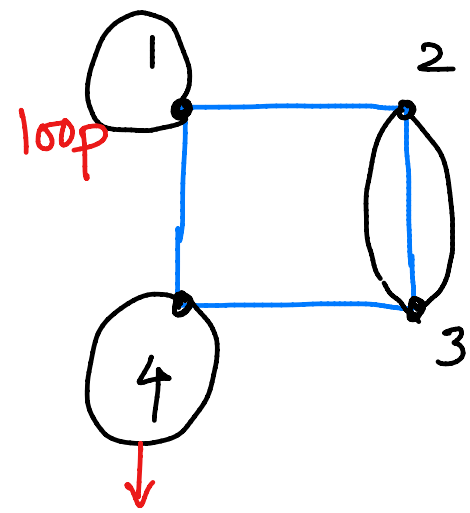
Graph

$G(V, E)$ is a graph consist of vertices and edges.

Vertex, Edge

edge is a connector between two vertices.





Vertex = 4

edges = 8

Simple graph: A graph without loop

any loop.

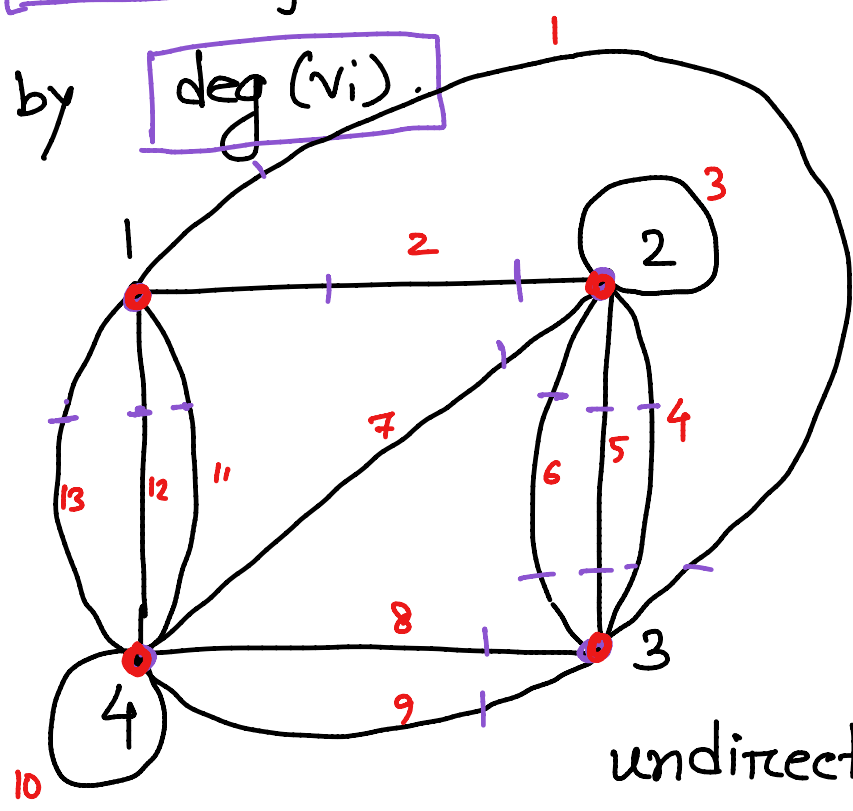
A graph with loop is call pseudograph.

Define degree of a graph and state Handshaking theorem.

Soln: Degree: consider a graph $G(V_i, E_i)$. The degree of vertex v_i means number of edges connected to the vertex v_i, denoted by $\deg(v_i)$.

Consider a graph

#note: A loop contribute 2 degree of a vertex.



undirected graph

$\deg(1) = 5$

$\deg(2) = 7$

$\deg(3) = 6$

$\deg(4) = 8$

$\Sigma \deg(v_i) = 26$

Hand-shaking Theorem: $2e = \Sigma \deg(v_i)$

Here, e = number of edges.

From the graph, $\Sigma \deg(v_i) = 26$

$2e = 26$

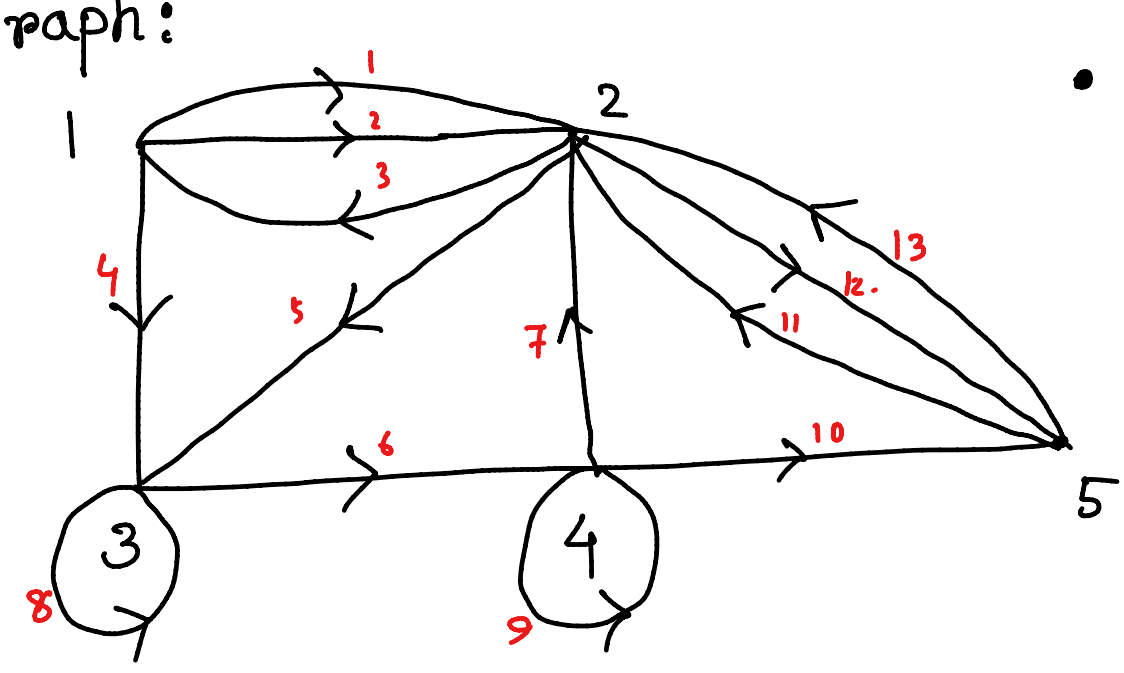
$\therefore e = 13$

From the graph, there are 13 edges.

∴ Hand shaking theorem varified.

pendent

Directed graph:



Total number of vertex = 6

egdes = 13

out degree $(\deg^+(v_i))$

in degree $(\deg^-(v_i))$

$\deg^+(1) = 3$ $\deg^-(1) = 1$

$\deg^+(2) = 3$ $\deg^-(2) = 5$

$\deg^+(3) = 2$ $\deg^-(3) = 3$

$\deg^+(4) = 3$ $\deg^-(4) = 2$

$\deg^+(5) = 2$ $\deg^-(5) = 2$

$\deg^+(6) = 0$ $\deg^-(6) = 0$

$\Sigma \deg^+(v_i) = 13$ $\Sigma \deg^-(v_i) = 13$

Hand-shaking theorem for directed graph

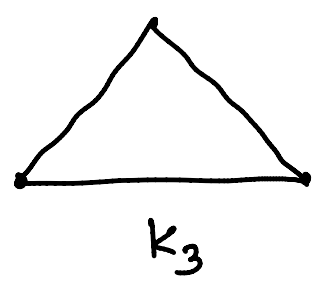
$$e = \Sigma \deg^+(v_i) = \Sigma \deg^-(v_i)$$

Different types of graph:

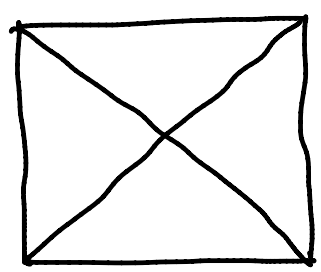
1. complete graph: (K_i)



K_2

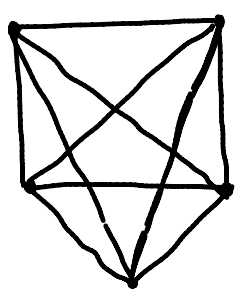


K_3

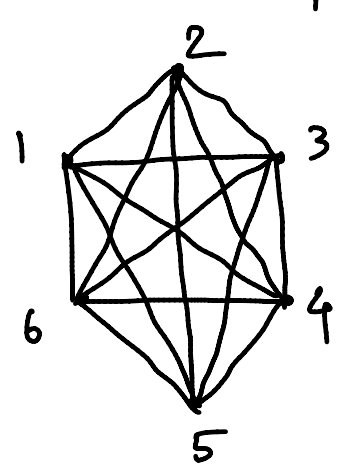


K_4

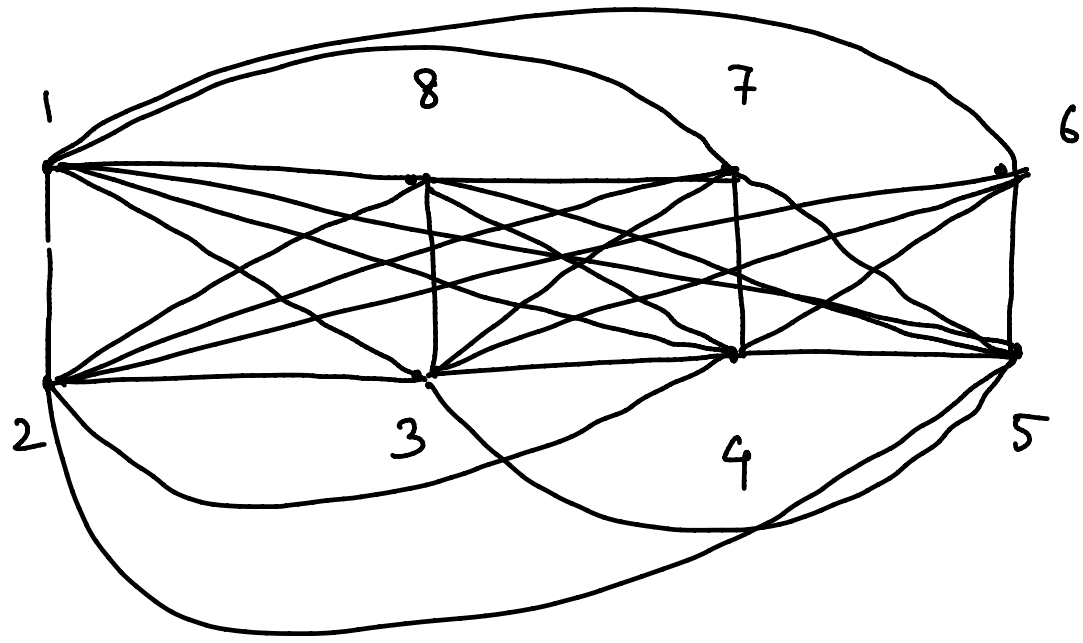
draw K_5



K_6:

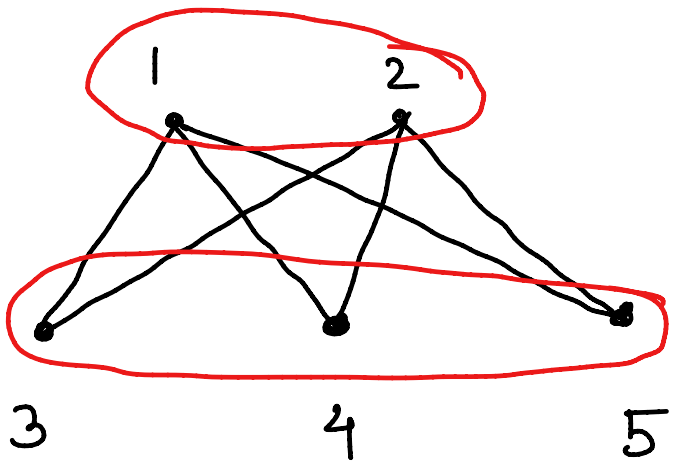


K_8:

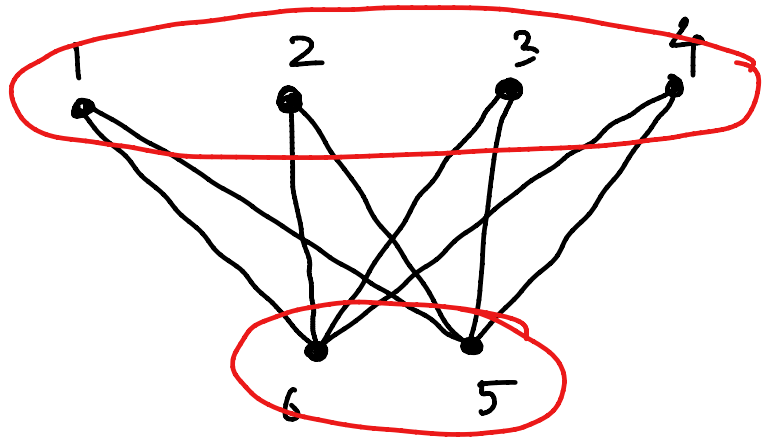


complete bipartite: $(K_{m,n})$

$K_{2,3}$



$K_{4,2}$



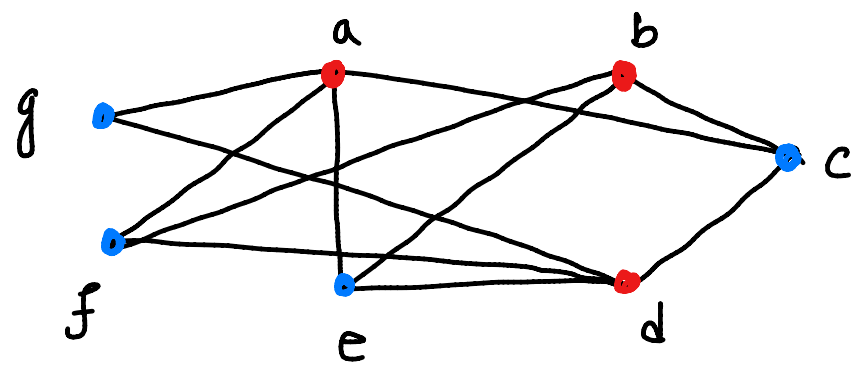
Determine

$$u = 1582(1 - e^{-y})$$

$$\frac{\partial u}{\partial y} = 1582\, e^{-y}$$

$$t = \mu \frac{\partial u}{\partial y}$$

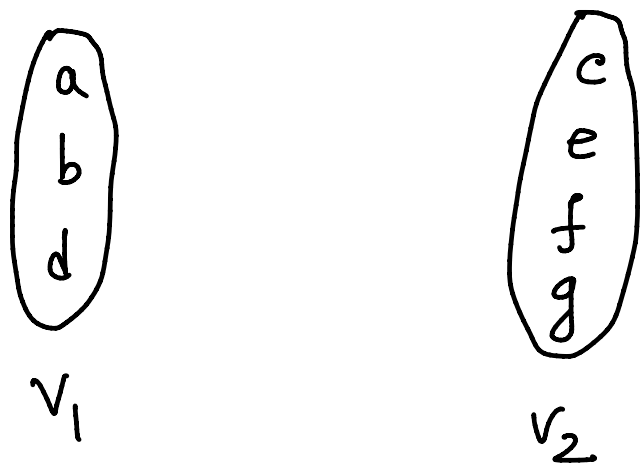
$$= \square \times 1582 e^{-0}$$



Is this graph bipartite or not ? ——— [7]

Soln: To determine whether the above graph bipartite or not we use two color theorem. Choose one is blue another is red.

Without loss of generality, choose a as a red. Now c, e, f, g is directly connected to a. so mark them as blue. Now a, b, d is adjacent of f mark them as red.



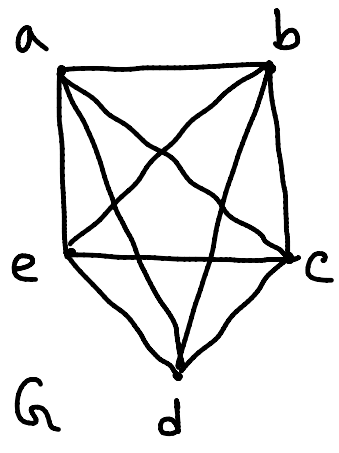
V_1 V_2

$V_1 \cap V_2 = \phi$

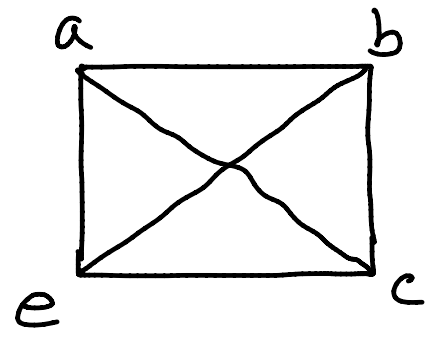
$\therefore$ so, G is bipartite.

Ans

Draw K_5 and Determine two subgraph.

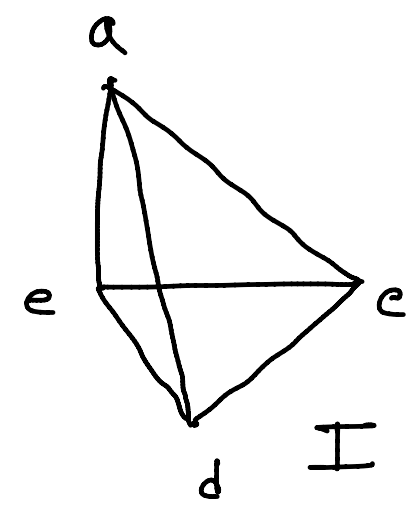


K_5



H

H ⊂ G



I

I ⊂ G

1. Mathematical Induction:

i) Use mathematical Induction to show that $n < 2^n$ for all positive integers n.

ii) using Mathematical induction to show that $1^2+3^2+5^2+\cdots+(2n+1)^2 = \frac{1}{3}(n+1)(2n+1)(2n+3)$.

for all ~~positive~~ non-negative integer.

Soln:

(i) Let $p(n): n < 2^n$

$n = 1$, $1 < 2^1$

$\therefore p(n)$ is true for $n = 1$.

Assume that $p(n)$ is true for $n = m$.

$$m < 2^m \quad \text{——— (i)}$$

Now we have to show that $p(n)$ is true for $n = m+1$

that is $m+1 < 2^{m+1}$ ——— (ii)

(i) $+1 \Rightarrow$ $m+1 < 2^m + 1$

$m+1 < 2^m + 1 < 2^m + 2 < 2 \cdot 2^m$

$m+1 < 2 \cdot 2^m$

$m+1 < 2^{m+1}$

$\therefore p(n)$ is true for $n = m+1$

$P(n)$ is true for all integers.

(proved)

$$\# \; a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(r^n - 1)}{r-1} \quad |r| > 1$$

$$= \frac{a(1-r^n)}{1-r} \quad |r| < 1$$

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} = \frac{1 - \frac{1}{2^n}}{\frac{1}{2}} = \frac{1}{2}\left(\frac{2^n - 1}{2^n}\right)$$

$$= \frac{2^n - 1}{2^{n+1}}$$

\# Htw: $1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} = \frac{2^n - 1}{2^{n+1}}$ for all positive integer.

(ii) Let $P(n)$: $1^2 + 3^2 + 5^2 + \cdots + (2n+1)^2 = \frac{1}{3}(n+1)(2n+1)(2n+3)$

$$\sum_{m=0}^{n} (2m+1)^2 = \frac{1}{3}(n+1)(2n+1)(2n+3)$$

1st step:

$$L.H.S = 1^2$$

$$R.H.S = \frac{1}{3}(0+1)(0+1)(0+3) = 1$$

$\therefore$ $P(0)$ is true or. $p(n)$ is true for $n = 0$.

$n = 1$,

$L.H.S: \; 1^2 + 3^2 = 10$

$R.H.S: \; \frac{1}{3} \cdot 255. = 10$

X

Assume that, $P(n)$ is true for $n = m$

$$1^2 + 3^2 + 5^2 + \cdots + (2m+1)^2 = \frac{1}{3}(m+1)(2m+1)(2m+3)$$

—— (i)

Now we have to prove that, $p(n)$ is true for $n = m+1$

$$1^2 + 3^2 + 5^2 + \cdots + (2m+1)^2 + (2(m+1)+1)^2 = \frac{1}{3}(m+1+1)(2(m+1)+1)(2(m+1)+3)$$

$$\underline{1^2+3^2+5^2+\cdots+(2m+1)^2} + (2m+3)^2 = \frac{1}{3}(m+2)(2m+3)(2m+5) \quad \text{— (ii)}$$

(i) + $(2m+3)^2$

$$1^2+3^2+5^2+\cdots+(2m+1)^2+(2m+3)^2 = \frac{1}{3}(m+1)(2m+1)(2m+3) + (2m+3)^2$$

$$= (2m+3)\left(\frac{1}{3}(m+1)(2m+1) + (2m+3)\right)$$

$$= (2m+3)\ \frac{(m+1)(2m+1) + 6m + 9}{3}$$

$$= (2m+3)\ \frac{2m^2 + m + 2m + 1 + 6m + 9}{3}$$

$$= (2m+3)\ \frac{2m^2 + 9m + 10}{3}$$

$$= (2m+3)\ \frac{2m^2 + 4m + 5m + 10}{3}$$

$$= (2m+3)\ \frac{2m(m+2) + 5(m+2)}{3}$$

$$= (2m+3)\ \frac{(m+2)(2m+5)}{3}$$

$$= \frac{1}{3}(m+2)(2m+3)(2m+5)$$

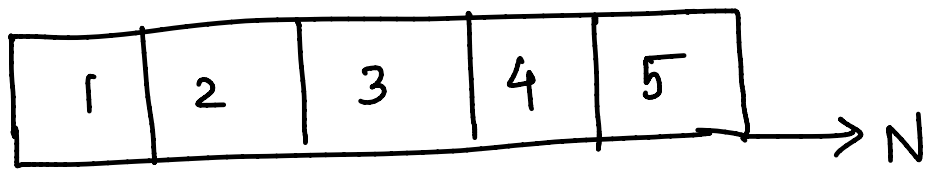
$p(n)$ is true for $n = m+1$

$p(n)$ is true for all non-negative integer.

The pigeonhole theorem:

books = 6

N+1



8

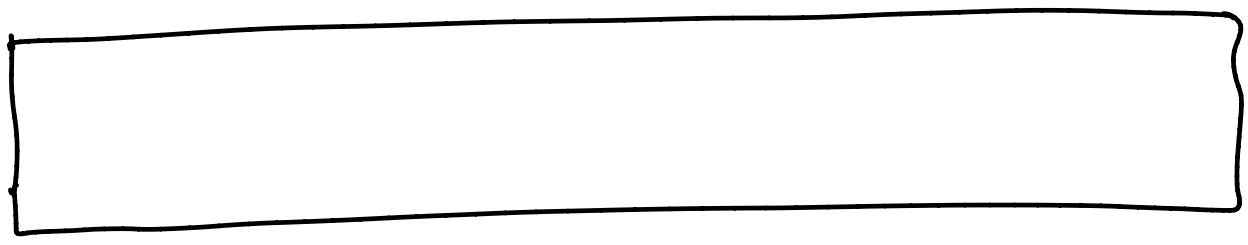
$$\left\lceil \frac{6}{5} \right\rceil = \lceil 1.2 \rceil = 2$$

11

$$\left\lceil \frac{11}{5} \right\rceil = \lceil 2.2 \rceil = 3$$

A+, A, B+, B, C+, C, D+, D, F (9 types)

Total students = 34



$$\left\lceil \frac{34}{9} \right\rceil = \lceil 3.7 \rceil = 4$$

$\left\lceil \frac{N}{9} \right\rceil = 10$

$\left\lceil \frac{89}{9} \right\rceil$

$N = 9 \times 9 + 1$

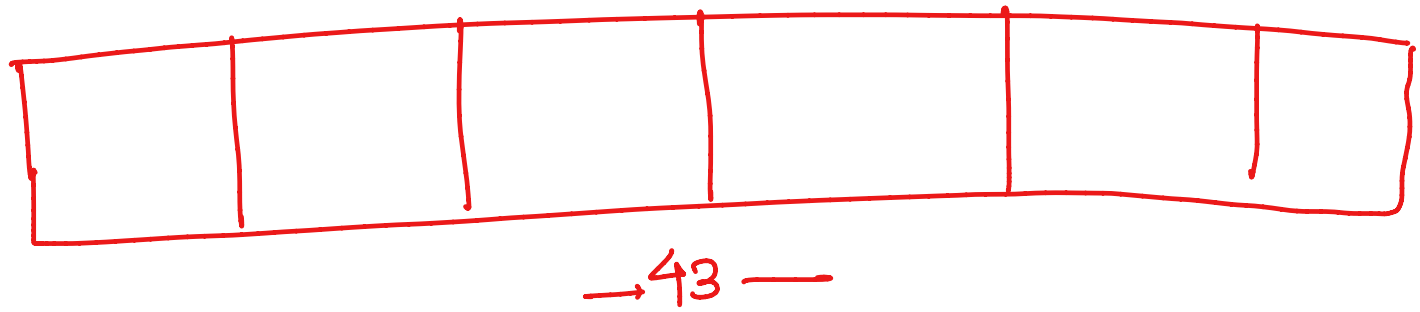
$= 82$

✓

There are 43 different time periods during which classes at a University can be scheduled. If there are 677 different classes, how many rooms will be needed?

↓ object

Soln:



$$\left\lceil \frac{677}{43} \right\rceil = \lceil 15.7 \rceil = 16$$

There are 40 students and minimum n 4 have the same grades. How many grades are possible.

Soln:

$N = 40$

$K = ?$

$$\left\lceil \frac{N}{K} \right\rceil = 4$$

$$\Rightarrow \left\lceil \frac{40}{K} \right\rceil = 4$$

minimum value of $K = 10$

10 grades

maximum value of $K = 13$

13 grades.

Boolean Dot product: $A \odot B$

Find Boolean dot product of $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}_{3\times2}$ and

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}_{3\times3}.$$

Soln:

$A_{3\times2}$ $B_{3\times3}$ = not possible

But $B_{3\times3}$ $A_{3\times2}$ $= BA_{3\times2}$

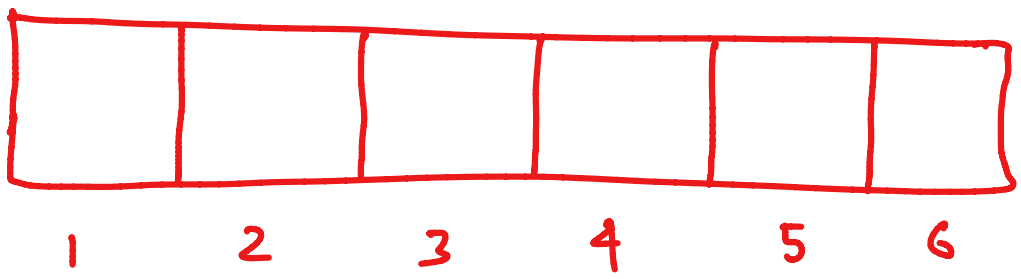
$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1\wedge1)\vee(0\wedge0)\vee(1\wedge1) & (1\wedge0)\vee(0\wedge1)\vee(1\wedge1) \\ (1\wedge1)\vee(1\wedge0)\vee(0\wedge1) & (1\wedge0)\vee(1\wedge1)\vee(0\wedge1) \\ (0\wedge1)\vee(1\wedge0)\vee(0\wedge1) & (0\wedge0)\vee(1\wedge1)\vee(0\wedge1) \end{bmatrix}$$

$$= \begin{bmatrix} 1\vee0\vee1 & 0\vee0\vee1 \\ 1\vee0\vee0 & 0\vee1\vee0 \\ 0\vee0\vee0 & 0\vee1\vee0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$ Ans

pigeonhole theorem:

Pigeon → 7



If there is N boxes and N+1 objects. Then at least one box contains more than one object.

There are 5 different time slots.

Number of classes = 287.

How many room will be needed?

→ At least, required number of room is = $\lceil \frac{N}{k} \rceil$

$= \lceil \frac{287}{5} \rceil$

$= \lceil 57.4 \rceil = 58$ Ans

Total student = 37

Total grades = 9 (A+, A, B+, B, C+, C, D+, D, F)

How many student get same grades?

$\lceil \frac{37}{9} \rceil = \lceil 4.11 \rceil = 5$

There are 5 possible grades. In a class how many lowest possible number of students are required for at least 4 students have grade?

$\lceil \frac{N}{K} \rceil = 4$ ✓ $\lceil \frac{N}{5} \rceil = 4$

$\lceil \frac{N}{5} \rceil = 4$

$\frac{N}{5} \leq 4$ and $\frac{N}{5} > 3$

$N > 15$

$N = 15+1 = 16$

Sunday

Quiz 1 → 1. Mathematical Induction & Boolean product

(30) 2. Pigeonhole theorem & Relations.

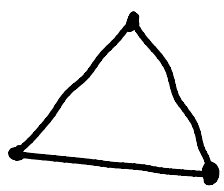
3. Basic graph (up to handshaking)

0 degree – isolated vertex
1 " → pendent "

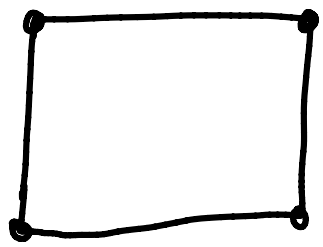
special types of graph:

1. complete graph: (K_n) . $n \geq 1$

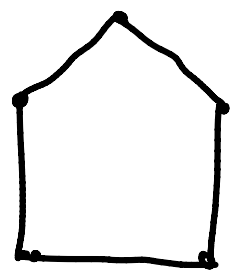
2. Cycles: C_n , $n \geq 3$



C_3

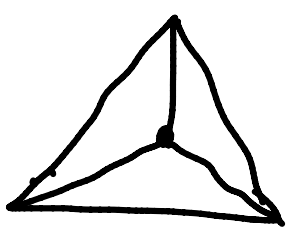


C_4

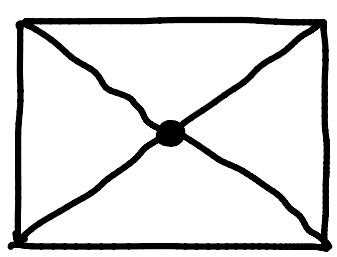


C_5

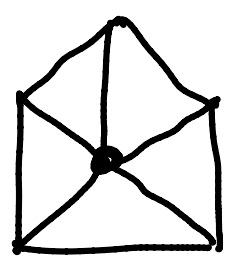
3. Wheels: W_n



W_3



W_4



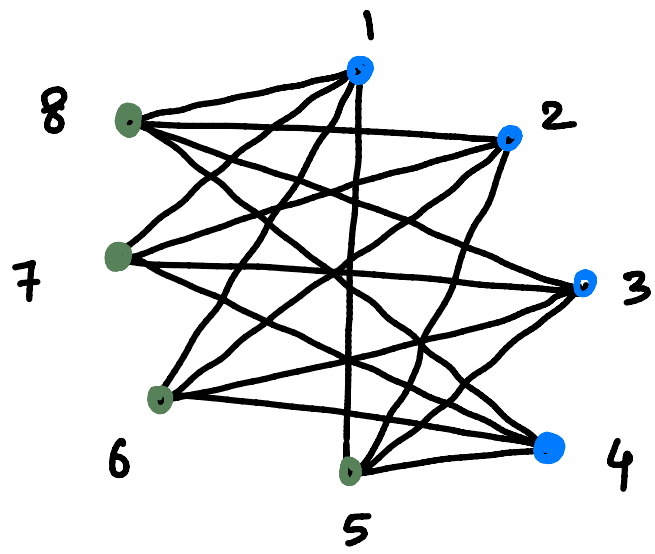
W_5

4. Biparctite graph: A graph (V, E) is said to be biparctite if it can be divided into two disjoints sets that (V_1, E_1) & (V_2, E_2) where

$$V_1 \cap V_2 = \phi.$$

two color theorem: with this theorem we can identify whether the graph is biparctite or not.

Qs. What is biparctite graph. Determine whether the following graph is biparctite or not?



Soln:

Choose two color. one is green and another blue.

Arcbitary, choose 7 as a green. directly adjacent to 7 are {1, 2, 3, 4} choose them as a blue. Now, the vertex {5, 6, 8} are adjacent to 2. mark them as a green.

Now form the color graph, we can conclude that. each edge's have different color vertex. so. It's a bipartite graph

Induction, Boolean product and pigeonhole theorem:

#1 $1+2+\cdots+n = \frac{n(n+1)}{2}$, $n \in N$

2. $1^2+2^2+\cdots+n^2 = \frac{1}{6}n(n+1)(2n+1)$, $n \in N$

3. $1^3+2^3+\cdots+n^3 = \left\{\frac{n(n+1)}{2}\right\}^2$, $n \in N$

4. $\frac{1}{1\cdot2} + \frac{1}{2\cdot3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$, $n \in N$

5. $1\cdot3 + 2\cdot3^2 + 3\cdot3^3 + \cdots + n\cdot3^n = \frac{(2n-1)3^{n+1}+3}{4}$. $n \in N$

Soln:

3. Let $P(n) = 1^3+2^3+\cdots+n^3 = \left\{\frac{n(n+1)}{2}\right\}^2$

1st step: $P(n)$ is true for $n=1$. because

$$L.H.S = 1^3 = 1$$

$$R.H.S = \left(\frac{1\cdot2}{2}\right)^2 = 1$$

2nd step: Let $P(n)$ is true for $n=m$

$$1^3+2^3+3^3+\cdots+m^3 = \left\{\frac{m(m+1)}{2}\right\}^2 \quad \text{---(i)}$$

3rd step: we have to show that, $P(n)$ is true for $n=m+1$

$$1^3+2^3+3^3+\cdots+m^3+(m+1)^3 = \left\{\frac{(m+1)(m+2)}{2}\right\}^2 \quad \text{---(ii)}$$

(i) + $(m+1)^3$

$$1^3+2^3+3^3+\cdots+m^3+(m+1)^3 = \left\{\frac{m(m+1)}{2}\right\}^2 + (m+1)^3$$

$$= \frac{m^2(m+1)^2}{4} + (m+1)^3$$

$$= \frac{m^2(m+1)^2 + 4(m+1)^3}{4}$$

$$= \frac{(m+1)^2\,(m^2 + 4(m+1))}{4}$$

$$= \frac{(m+1)^2\,(m^2+4m+4)}{4}$$

$$= \frac{(m+1)^2\,(m+2)^2}{2^2} = \left\{\frac{(m+1)(m+2)}{2}\right\}^2$$

$\therefore$ $p(n)$ is true for $n = m+1$.

⑤ $p(n) = 1\cdot3 + 2\cdot3^2 + 3\cdot3^3 + \cdots + \underline{n\cdot3^n} = \dfrac{(2n-1)\,3^{n+1}+3}{4}$ because

<u>1st step</u>: $p(n)$ is true for $n=1$

$$L.H.S = 1\cdot3^1 = 3$$

$$R.H.S = \frac{(2-1)\,3^2+3}{4} = 3$$

<u>2nd step</u>: Assume that, $p(n)$ is true for $n=m$

$$1\cdot3 + 2\cdot3^2 + 3\cdot3^3 + \cdots + m\cdot3^m = \frac{(2m-1)\,3^{m+1}+3}{4} \quad \text{— (i)}$$

<u>3rd step</u>: We have to show that, $p(n)$ is true for $n = m+1$

$$1\cdot3 + 2\cdot3 + \cdots + m\cdot3^m + (m+1)\,3^{m+1} = \frac{(2m+1)\,3^{m+2}+3}{4} \quad \text{— (ii)}$$

(i) $+ (m+1)\cdot3^{m+1}$

$$1\cdot3 + 2\cdot3^2 + 3\cdot3^3 + \cdots + m\cdot3^m + (m+1)\cdot3^{m+1} = \frac{(2m-1)\,3^{m+1}+3}{4} + (m+1)\cdot3^{m+1}$$

$$= \frac{(2m-1)\,3^{m+1}+3+4(m+1)\,3^{m+1}}{4}$$

$$= \frac{3^{m+1}(2m-1+4m+4)+3}{4}$$

$$= \frac{3^{m+1}\cdot(6m+3)+3}{4}$$

$$= \frac{3\cdot 3^{m+1}(2m+1)+3}{4}$$

$$= \frac{(2m+1)3^{m+2}+3}{4}$$

$\therefore P(n)$ is true for $n=m+1$

Ans.

Boolean product:

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$R = \{(1,1), (1,2), (1,3), (2,1), (2,3), (2,4), (3,1), (3,3), (4,3), (4,4)\}$$

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

R is not Reflexive

not symmetric ✓

not Anti symmetric ✓

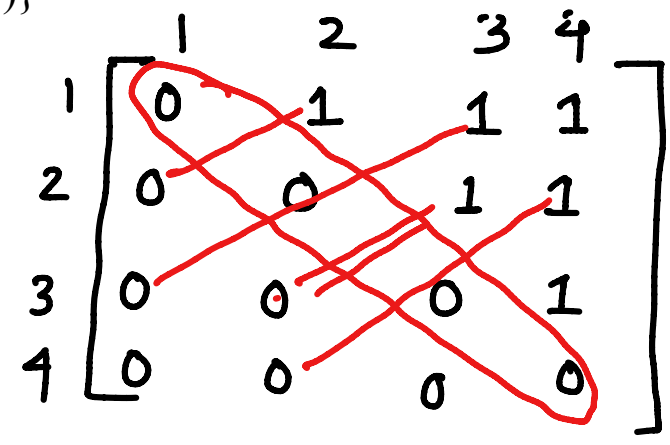
$$M_R \odot M_R = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

not transitive $= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

Let A= {1, 2, 3, 4}, determine whether the relations are reflexive, symmetric, anti-symmetric or transitive. R= {(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)}

Soln: **Relation matrix:**

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$



Graphical

$$M_R \odot M_R = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

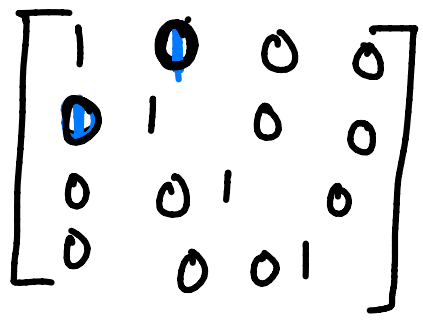
$$M_R^2 = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

R is not reflexive.

R is not symmetric

Anti-symmetric.

transitive



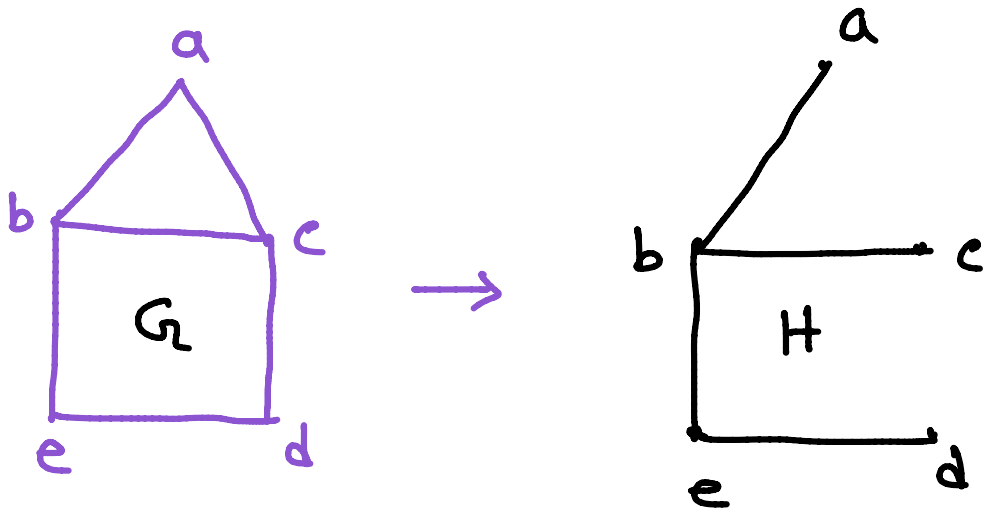
Subgraph:

Subgraph: H is called subgraph of G_2 if

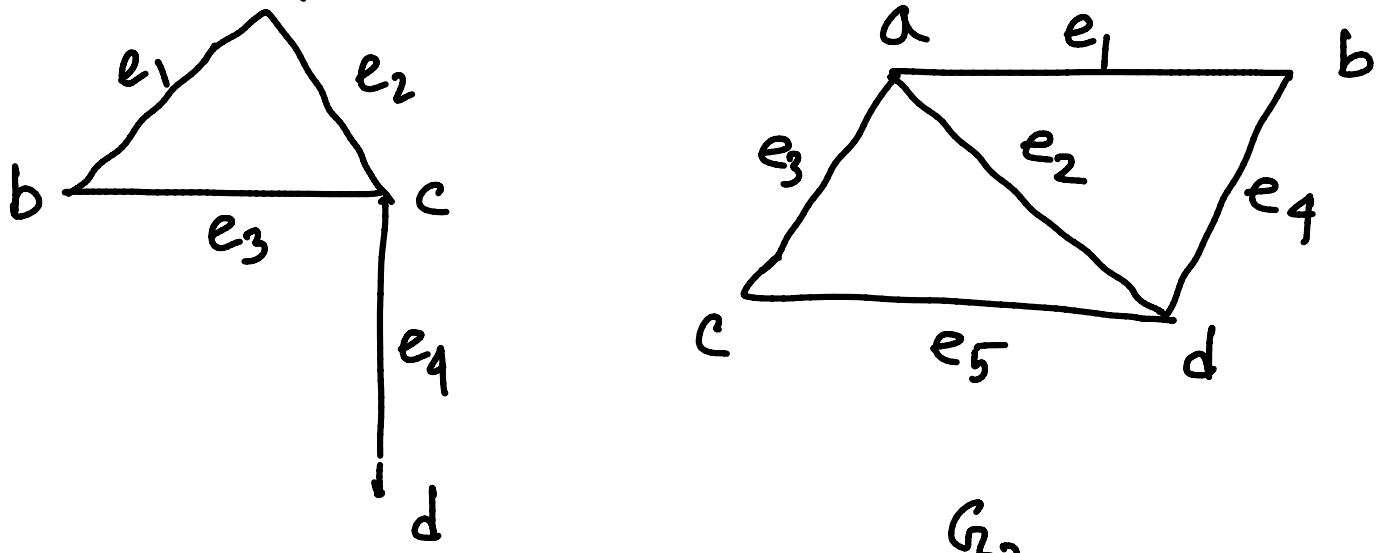
(i) All the verticies of H are in G_2

(ii) All edges of H are in G_2

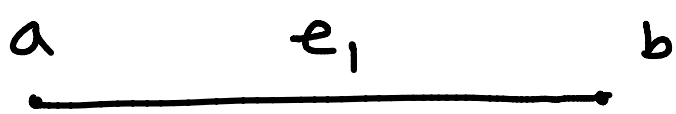
(iii) Same end points.



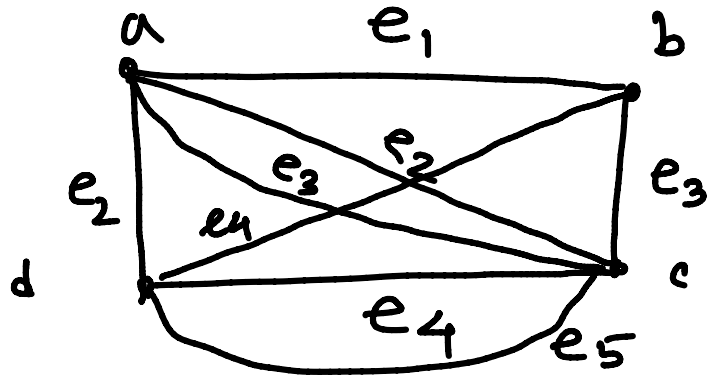
(b) $G_1 \cap G_2$



Find $G_1 \cap G_2$



Find $G_1 \cup G_2$:



Draw the graph using incidence matrix :

$$G_2 = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

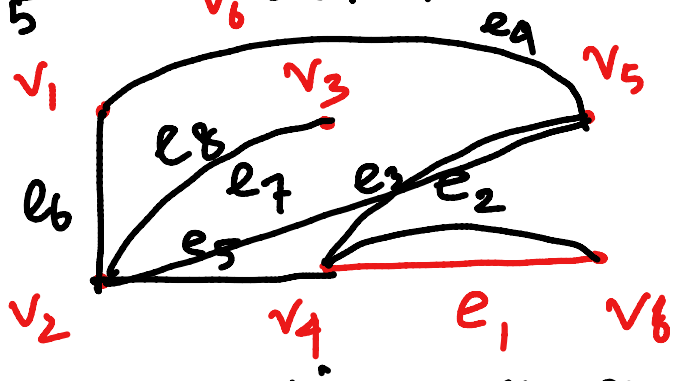
5x5

$$H = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

6x8

Vertex = 5

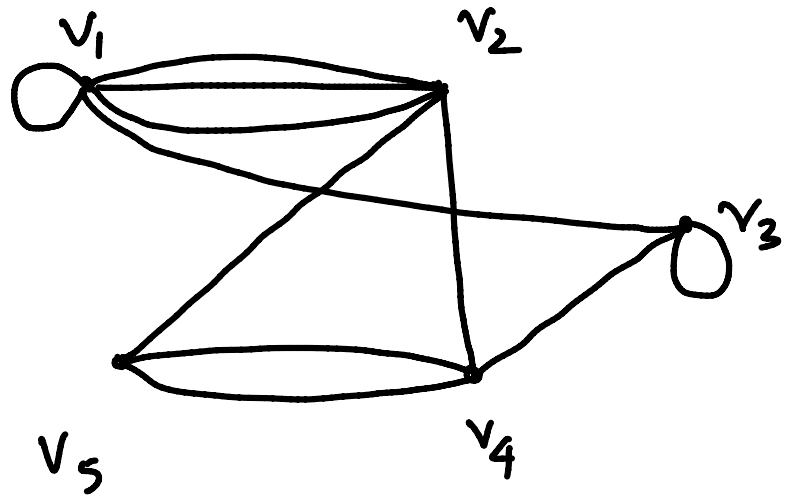
edge = 5



Adjacency matrix

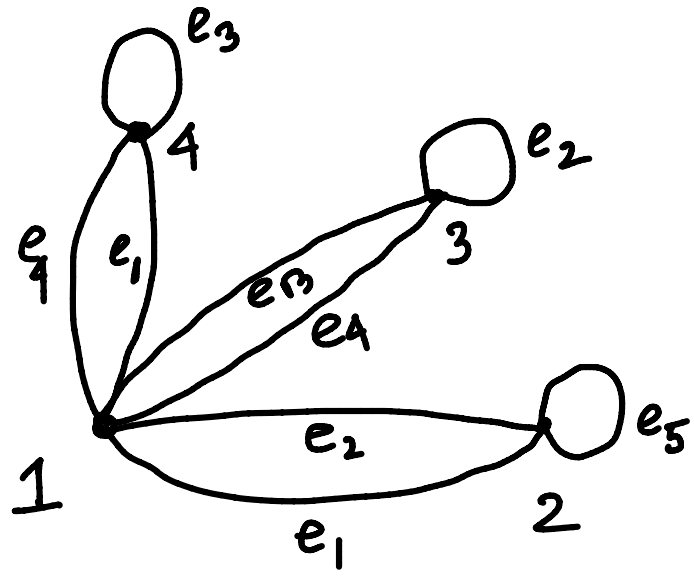
Draw the graph using

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_3 & v_4 & v_5 \end{array} \\ \begin{bmatrix} 1 & 3 & 1 & 0 & 0 \\ 3 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 2 & 0 \end{bmatrix} \end{array}$$



A

From the graph below find Adjacent matrix & Incident matrix.



G_1

Soln: It's an undirected graph.

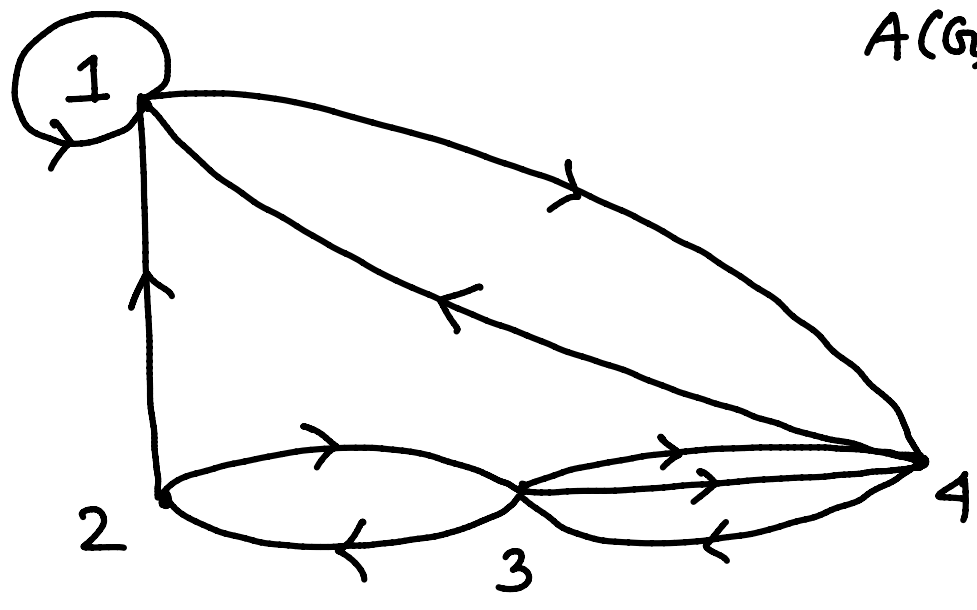
Now, Adjacent matrix is a matrix of vertex to vertex.

$$A(G_1)=\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Incident matrix is a matrix of vertex to edges

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{array} = I(G_1)$$

2. Find Adjacent matrix from the directed graph.

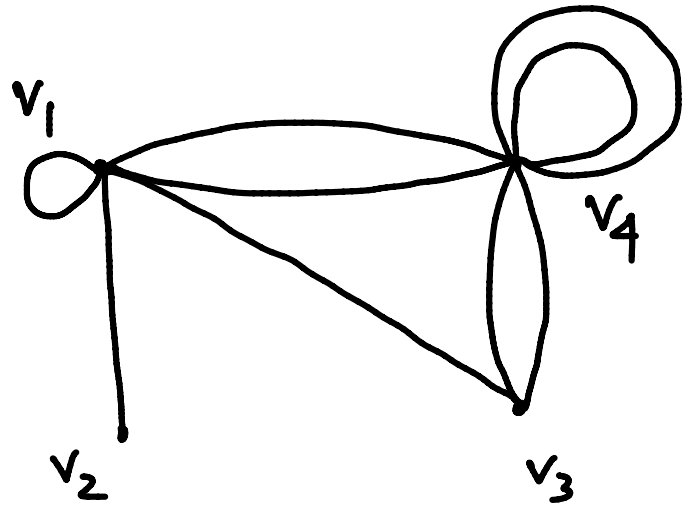


$$A(G_1)=\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Find the graph from the following matrix.

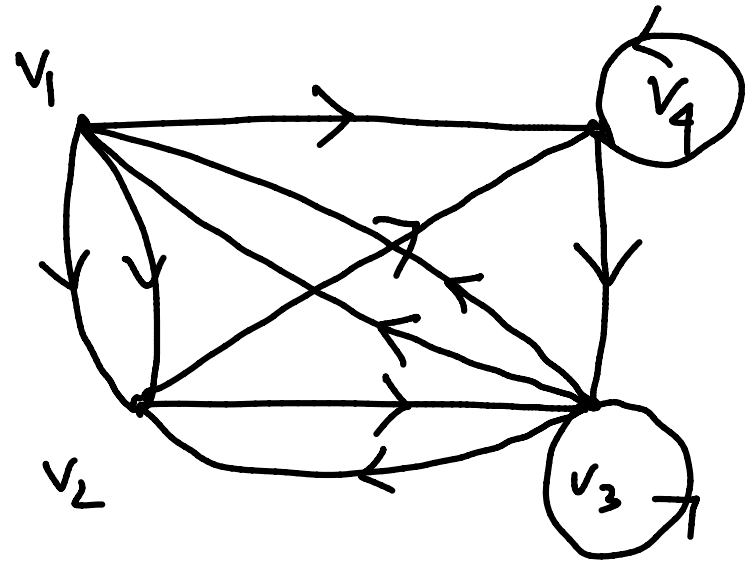
	v_1	v_2	v_3	v_4
v_1	1	1	1	2
v_2	1	0	0	0
v_3	1	0	0	2
v_4	2	0	2	2

undirected



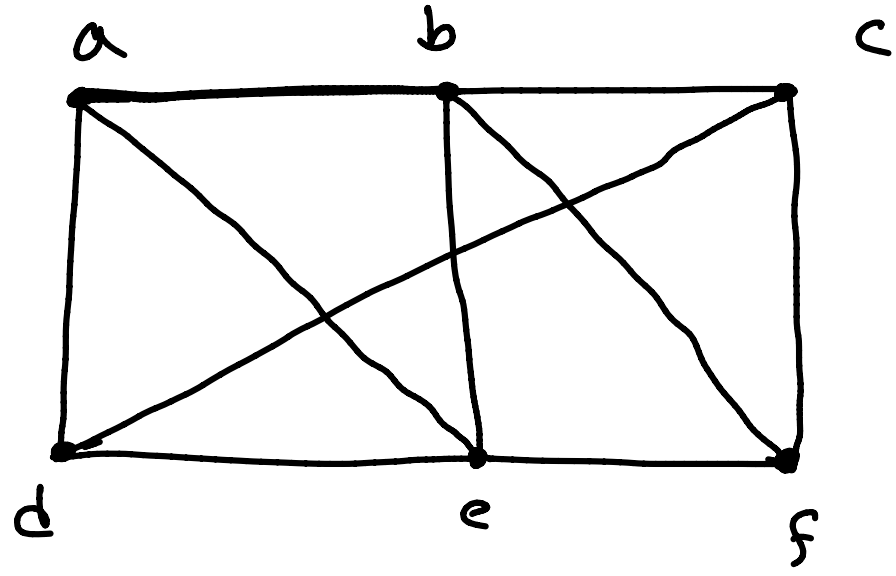
	v_1	v_2	v_3	v_4
v_1	0	2	0	1
v_2	0	0	1	1
v_3	2	1	1	0
v_4	0	0	1	1

directed

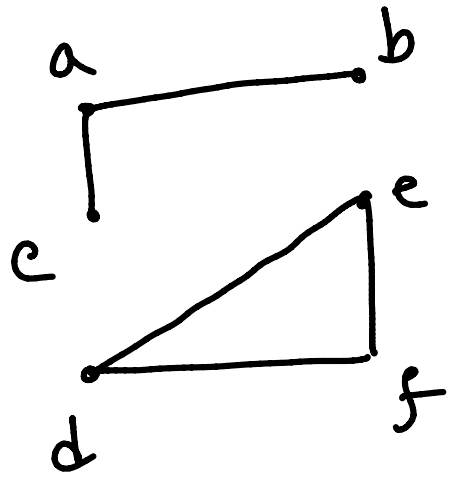


Connectivity

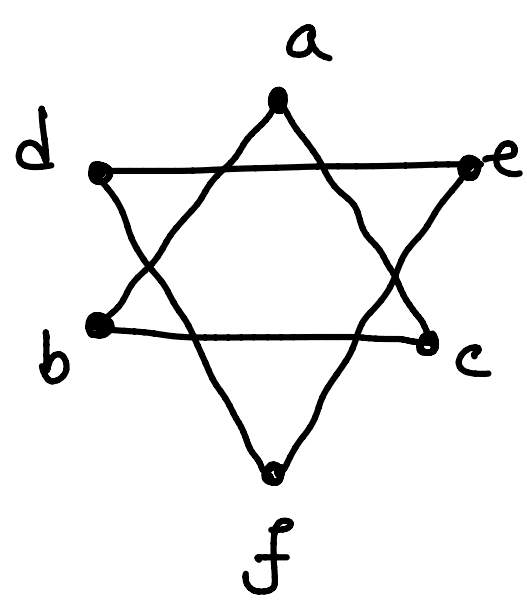
An undirected graph is called connected if there is a path between every pair of distinct vertices of the graph.



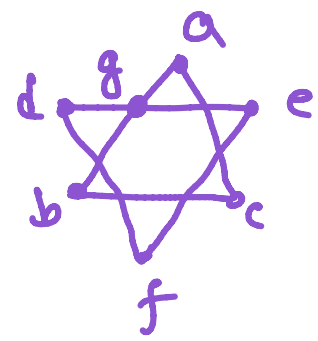
{ (a,d), (d, c), (f, b) } $a \to b$ length = 3



not connected



not connected



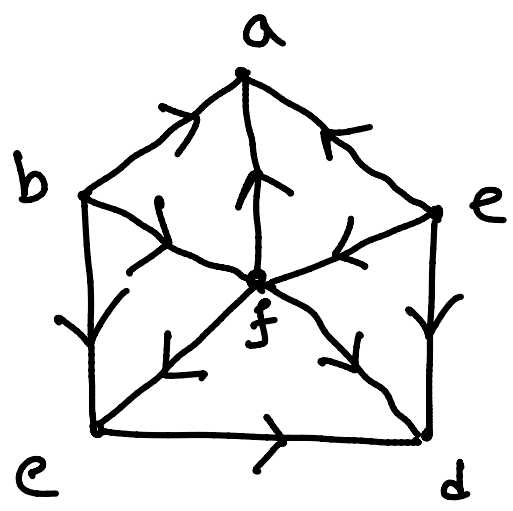
connected

Strongly & weakly connected graph:

Defⁿ:

A graph (directed) is ~~strong~~ strongly connected if there is a path from a to b and b to a.

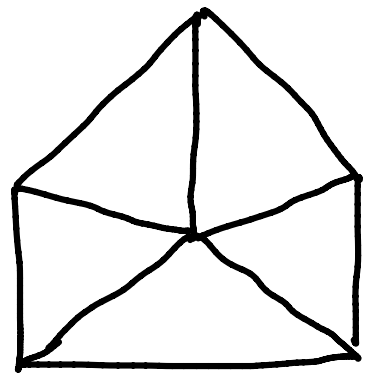
A directed graph is weakly connected if there is a path between every vertex.



→ There is no path from a to b.

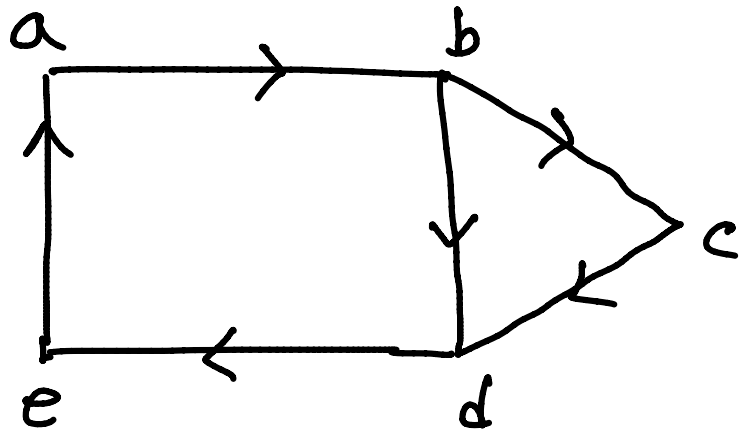
So the graph is not strogly connected.

weakly connected: (without any direction)



without direction. the graph is connected.

this is weakly connected.



Is the following graph is strongly connected or weakly connected or both ?

Soln:

a → b

a → b → c

a → b → d

a → b → d → e

b → d → e → a

b → c

b → d

b → d → e

c → d → e → a

c → d → e → a → b

c → d

c → e

d → e → a

d → e → a → b

d → e → a → b → c

d → e

e → a

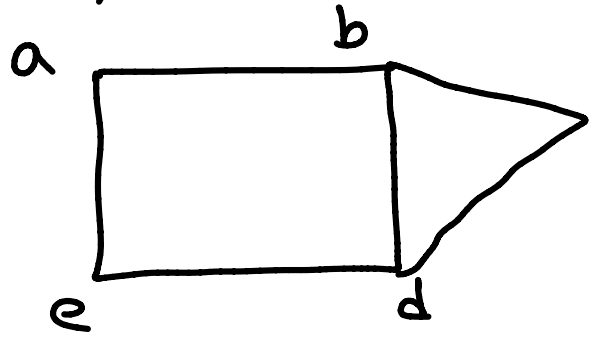
e → a → b

e → a → b → c

e → a → b → d

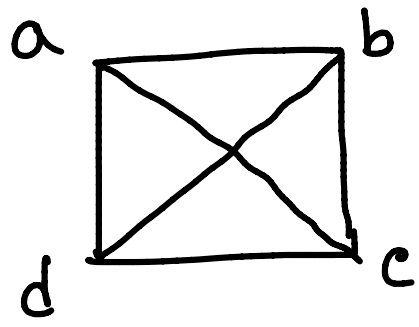
Strongly connected.

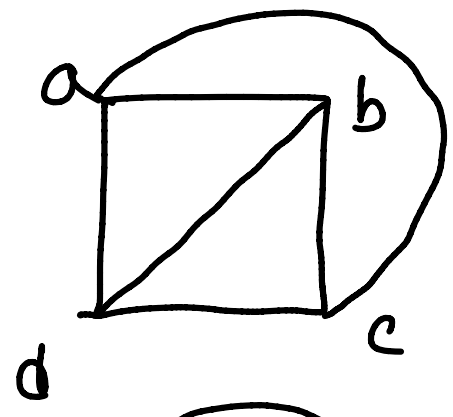
without any direction



weakly connected

Ans





planar graph

